PATHOLOGY OF BONES, JOINTS, MUCLES & SOFT-TISSUE TUMOURS

MBS 320

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LECTURE SERIES OBJECTIVE

- At the end of the lecture series, we should have discussed the following
- Review of the anatomy and Physiology of the bone
- Genetic, Metabolic bone diseases and bone fractures (BONES)
- Arthropathies (JOINTS)
- Musculopathies (MUSCLES)
- Tumours of the bone, cartilage, muscles and soft tissues

INTRODUCTION

- The skeletal system consists of bones and other structures that make up the joints of the skeleton.
- The types of tissue present are bone tissue, cartilage, and fibrous connective tissue, which forms the ligaments that connect bone to bone

CLASSIFICATION (TYPES) OF BONES

- According to position
  - AXIAL
  - APPENDICULAR
- According to Structure
  - COMPACT(CORTICAL)
  - CANCELLOUS (TRABECULAR OR SPONGY)
- According to development
  - INTRAMEMBRANOUS
  - INTRACARTILAGENOUS
- According to their shapes
  - LONG, SHORT, FLAT, IRREGULAR, ETC

FUNCTIONS OF THE SKELETON

1. Provides a framework that supports the body; the muscles that are attached to bones move the skeleton.
2. Protects some internal organs from mechanical injury; the rib cage protects the heart and lungs, for example.
3. Contains and protects the red bone marrow, the primary hemopoietic (blood-forming) tissue.
4. Provides a storage site for excess calcium+ PO_4. Calcium may be removed from bone to maintain a normal blood calcium level, which is essential for blood clotting and proper functioning of muscles and nerves.

Overview

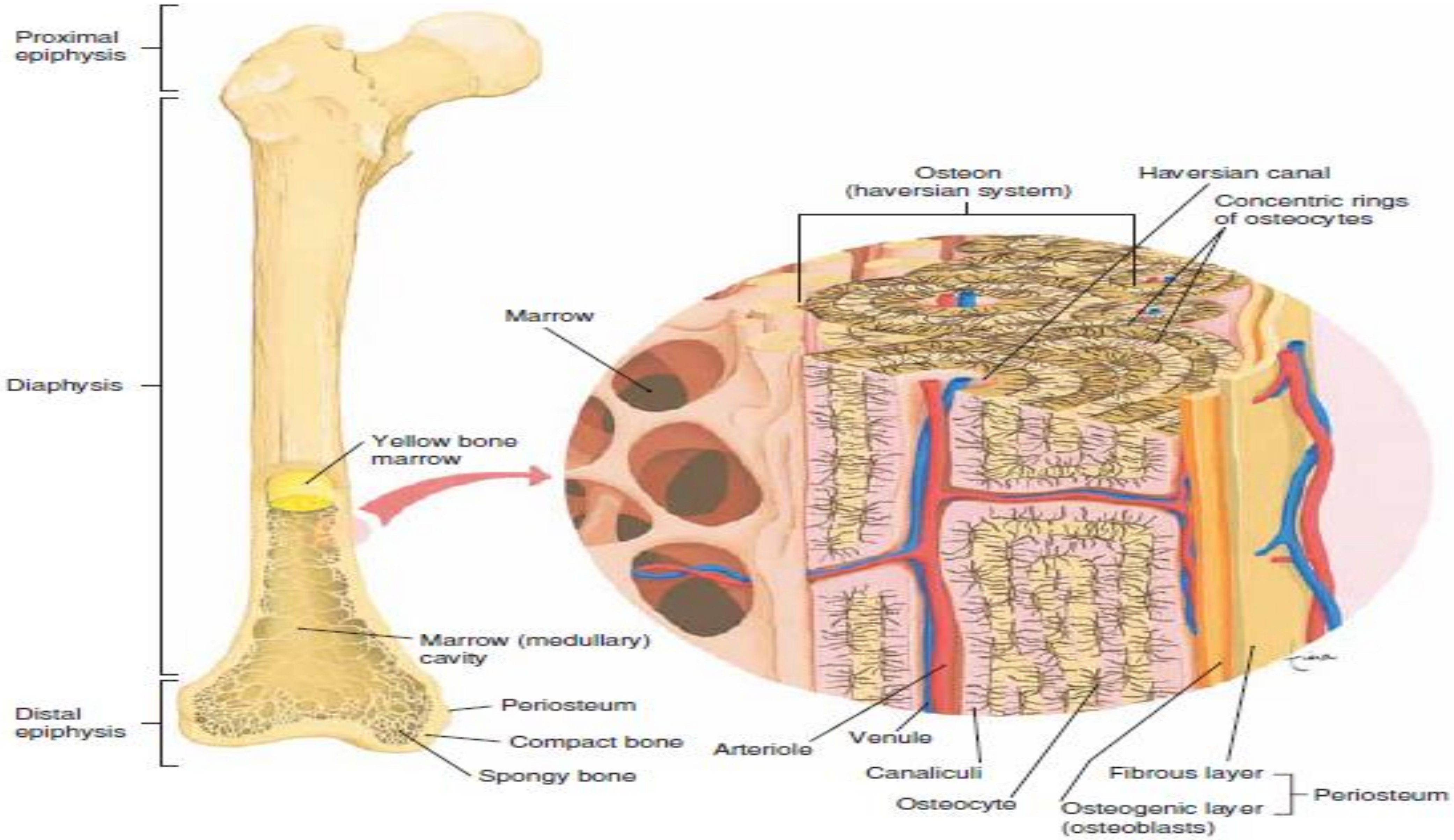
- Bone is non-living tissue composed of tissue bone cells are called osteocytes, and the matrix of bone is made of calcium salts and collagen (Matrix + Cells + Minerals)
- In normal circumstances, the amount of calcium that is removed is replaced by an equal amount of calcium deposited
- This is the function of osteocytes, to regulate the amount of calcium that is deposited in, or removed from, the bone matrix.
- The also regulate the function of Osteoclast and Osteoblasts by a process known as bone remodelling
    - Mature bone tissue is removed REABSORPTION
    - New bone tissue is laid/added through OSSIFICATION

Cont'd

- Bone is composed of specialized collagen-containing extracellular matrix (osteoid), which is mineralized by deposition of hydroxyapatite – osteoid is synthesized by osteoblast
- In normal lamellar bone, osteoid collagen is deposited in the mechanically strong, parallel stratified pattern.
- In abnormal woven bone, the osteoblasts is deposited in the haphazard pattern – has a tendency to fracture under stress
- Osteoclasts are highly specialised multinucleate cells derived from the Monocyte-Macrophage series capable of removing bone thus

Cont’d

- The control of osteoblast and osteoclast activity is crucial for normal bone physiology
- Majority of bone diseases are due either to acute trauma (such as fracture of the bone, rupture of the tendons and sprains of the joints) or to chronic degenerative diseases due to wear and tear (such as osteoarthrosis and the various wear and tear lesions of the tendons, tendon sheaths & ligament)
- Bone tumours are rare primary lesion, but the bones is an important & common site for metastasis



Proximal epiphysis
Diaphysis
Yellow bone marrow
Marrow (medullary) cavity
Distal epiphysis
Periosteum
Compact bone
Spongy bone
Marrow
Osteon (haversian system)
Haversian canal
Concentric rings of osteocytes
Arteriole
Venule
Canaliculi
Osteocyte
Fibrous layer
Osteogenic layer (osteoblasts)
Periosteum

Genetic Bone Diseases

Achondroplasia

- Is an autosomal dominant condition which is the most common cause dwarfism
- Results from a heterozygous mutation in the gene that code for FGFR3 on chromosome 4
- The pathogenesis results from the replacement of the glycine on AA380 with arginine which constitutively actives the FGFR3 leading to the inhibition of bone growth
- Normally, FGF-mediated activation of FGFR3 inhibits cartilage proliferation – so mutation affection endochondral bone formation (long bones such as the humerus, phalanges)

Cont’d

- Intramembranous bone formation is less affected & this is the growth with pre-existing cartilage matrix thus flat bone are not overtly affected (Skull, Ribs)

Features

- Results of shortened long bones is disproportionate short stature
- Shortened proximal extremities (rhizomellic), leg deformity (“kness out”), short metacarpals, short phalanges (brachydactyly)
- Other features include large head size, frontal bossing, flattened nasal bridge, narrow foramen magnum and spinal lordosis
- The skeletal changes are not associated with changes in longevity, intelligence or fertility

Thanatophoric dwarfism

- Is the most common lethal form of dwarfism characterised with a gain-of-function in the FGFR3
- Affected individual have micromelic shortening of the limbs, frontal bossing, relative microcephaly, a small chest cavity, a bell shaped abdomen
- Underdeveloped thoracic cavity leads to respiratory insufficiency and individuals usually die at birth or short after birth

Type 1 collagen disease (osteogenesis imperfecta)

- Is a phenotypically diverse disorder caused by deficiencies in the synthesis of collagen type 1
- It is also called brittle bone disease & is the most common inherited disorder of the connective tissue
- Is an autosomal dominant mutation in the genes that code for $\alpha 1$ & $\alpha 2$ chains of collagen
- Primarily it affects bone but also impacts type 1 rich collagen tissues such as skin, joints, eyes, ears, teeth

Cont'd

- **Blue sclerae** from translucency of thin connective tissue overlying the choroid due to decreased collagen are often present.
- The several clinical types vary greatly in severity.
- In the most common type, an autosomal dominant variant, blue sclerae, hearing loss and multiple childhood fractures are prominent clinical findings.

Osteopetrosis

- Is called marble disease & Albers-Schönberg disease.
- Can be autosomal dominant or recessive, Bones are brittle and fracture easily
- Is a rare group of genetic disorders that are characterised by reduced bone resorption and diffuse symmetrical skeletal sclerosis due to impaired formation or function of osteoclast
- increased skeletal mass or osteosclerosis caused by a hereditary defect in osteoclast function
- The pathogenesis mostly involve mutations that interfere with process of acidification of the osteoclast resorption pit, which is required for the dissolution of calcium hydroxyapatite e.g. Defects in the CA2 gene

Features

- Clinical features in autosomal dominant are evident in utero or soon after birth
- Fracture, anaemia & hydrocephaly
- Affected individuals who survive have cranial nerve defects, repeated infections, Hepatosplenamegaly
- In mild forms, may not be detected until adolescence or adult hood due to repeated fracture
- Can be cured with bone marrow transplant

Metabolic bone disease

Osteoporosis (porous bone)

- Is a group of progressive bone disease characterized by porous bones and reduced bone mass
- Results from imbalance between bone resorption and bone formation
- is most common metabolic bone disease
- Bone is mostly composed of thin trabecular which is particularly predisposed to fracture with minimal trauma & is widespread in the elderly as an important cause of morbidity and even mortality
- May be localised to a bone or region or may be generalized
- Osteoporosis can be primary such as postmenopausal, senile or idiopathic
- Or secondary due to endocrine disorders, neoplasia, gastrointestinal, drugs or miscellaneous

Epidemiology

- More common in Caucasians than Blacks
- More frequent in women than men especially post menopausal
- Incidence increases with age
- Peak bone density is reached at about 20 – 25 years of age
- Bone remodeling or bone turnover is balanced up to this period of time
- Thereafter a small deficit in bone formation accrues with every resorption

Generalised osteoporosis

- Primary
  - Postmenopausal
  - Senile
  - idiopathic
- Secondary
  - Endocrine disorders
  - Gastrointestinal conditions
  - Drugs
  - Others

Pathogenesis

- *Age-related changes* in bone cells and matrix have a strong impact on bone metabolism.
- Reduced proliferative and biosynthetic potential when compared with osteoblasts from younger individuals.
- Also, proteins bound to the extracellular matrix (such as growth factors, which are both mitogenic to osteoprogenitor cells and stimulate osteoblastic synthetic activity) lose their biologic punch over time. The net result is a diminished capacity to make bone.
- This form of osteoporosis, known as *senile osteoporosis*, is categorized as a *low-turnover variant*.

Cont’d

- *Reduced physical activity* increases the rate of bone loss in experimental animals and humans, because mechanical forces stimulate normal bone remodeling.
- The type of exercise is important, as load magnitude influences bone density more than the number of load cycles.
- Because muscle contraction is the dominant source of skeletal loading,
- resistance exercises such as weight training are more effective stimuli for increasing
- bone mass than repetitive endurance activities such as jogging.
- Certainly the decreased physical activity that is associated with aging contributes to senile osteoporosis.

Cont’d

- The body's *calcium nutritional state* is important.
- It has been shown that adolescent girls (but not boys) tend to have insufficient calcium intake in the diet.
- This calcium deficiency occurs during a period of rapid bone growth, stunting the peak bone mass ultimately achieved; thus, these individuals are at greater risk of developing osteoporosis.
- Calcium deficiency, increased PTH concentrations, and reduced levels of vitamin D may also have a role in the development of senile osteoporosis.

Cont’d

- *Genetic factors are also important,*
- It is estimated that 60% to 80% of the variation in bone density is genetically determined. In genome-wide association studies (GWAS), the top associated genes include *RANKL(Receptor Activator of Nuclear factor Kappa beta Ligand)*, *OPG (osteoprotegerin)*, and *RANK*, which you will recall encode key regulators of osteoclasts.
- Also associated are the MHC locus (perhaps reflecting the effects of inflammation on calcium metabolism) and the estrogen receptor gene Some studies have also implicated genetic variants of the vitamin D receptor and LRP5 as risk factors.

Cont’d

- *Hormonal influences*. In the decade after menopause, yearly reductions in bone mass may reach up to 2% of cortical bone and 9% of cancellous bone.
- It is thus no surprise that post-menopausal women suffer osteoporotic fractures more commonly than men of the same age.
- *Postmenopausal* osteoporosis is characterized by a hormone-dependent acceleration of bone loss that occurs during the decade after menopause.
- *Estrogen deficiency plays the major role in this phenomenon, and estrogen replacement at menopause is protective against bone loss*.
- The effects of estrogen on bone mass are mediated by cytokines.
- Decreased estrogen levels appear to result in increased secretion of inflammatory cytokines by blood monocytes and bone marrow cells.
- These cytokines stimulate osteoclast recruitment and activity by increasing the levels of RANKL while diminishing the expression of OPG.
- Compensatory osteoblastic activity occurs, but it does not keep pace, leading to what is classified as a *high-turnover form* of osteoporosis.

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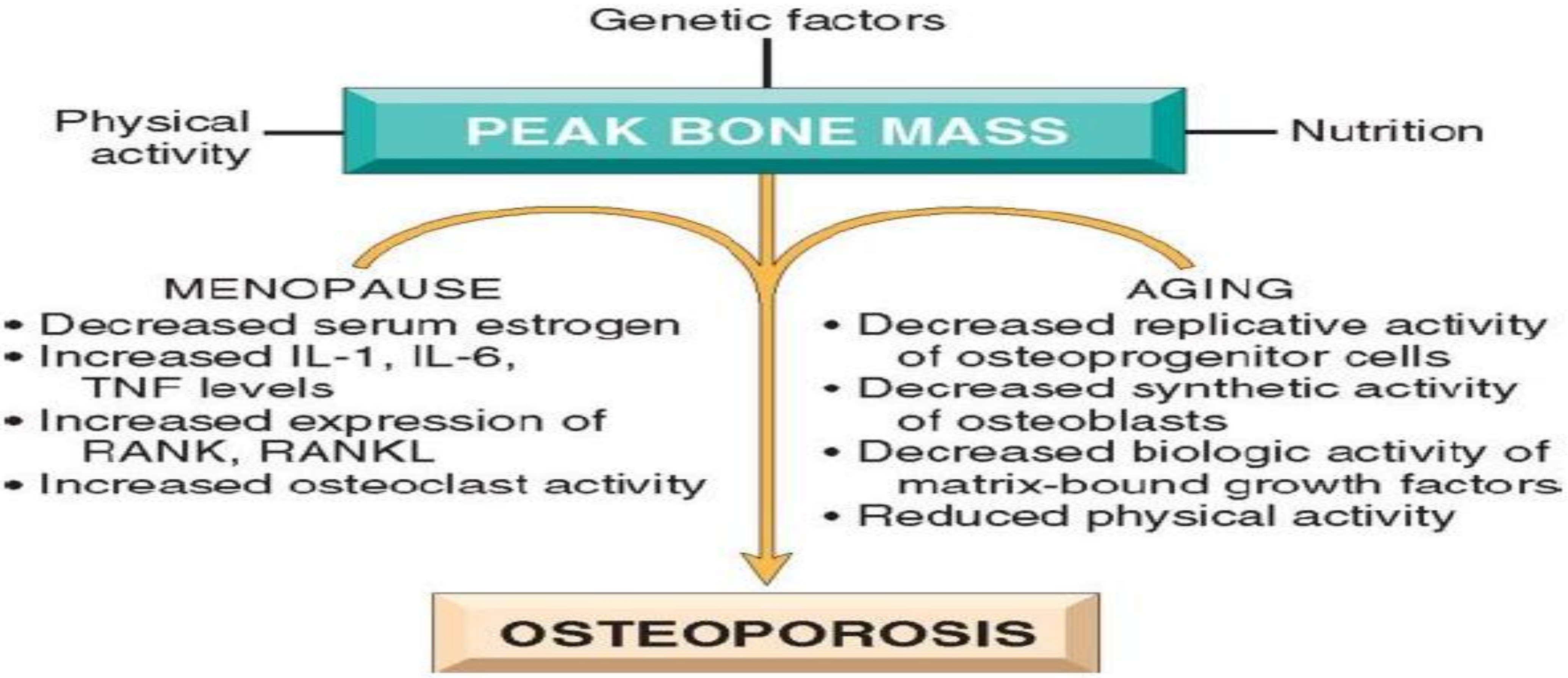

Genetic factors
Physical activity
PEAK BONE MASS
Nutrition
MENOPAUSE
• Decreased serum estrogen
• Increased IL-1, IL-6, TNF levels
• Increased expression of RANK, RANKL
• Increased osteoclast activity
AGING
• Decreased replicative activity of osteoprogenitor cells
• Decreased synthetic activity of osteoblasts
• Decreased biologic activity of matrix-bound growth factors
• Reduced physical activity
OSTEOPOROSIS

Clinical features/Risk factors

- Bone pain particularly in the back
- Multiple compression of the vertebrae can lead to overall loss of height – leading to bending of the spine anterioposterioly (kyphosis)
- Fractures in the neck of the femur & wrist(distal radius) are common complication in the elderly
- Age
- Low estrogen, females
- Low calcium
- Alcohol consumption & smoking
- Drugs
- Physical inactivity
- Diseases (hyperprolactinemia, diabetes mellitus, Cushing syndrome)

Diagnosis & Treatment

- Diagnosis is very difficult as osteoporosis remains asymptomatic until skeletal fragility is well advanced
- Conventional x-ray – requires 30% bone loss to be apparent
- MRI, CT
- Dual-energy X-ray absorptiometry
  - Measures bone density with a score of less than -2.5 on the T score been diagnostic
- Treatment includes exercise, appropriate calcium & vitamin D intake
- Bisphosphonates which inhibit such as osteoclast activity such as alendronate, risedronate

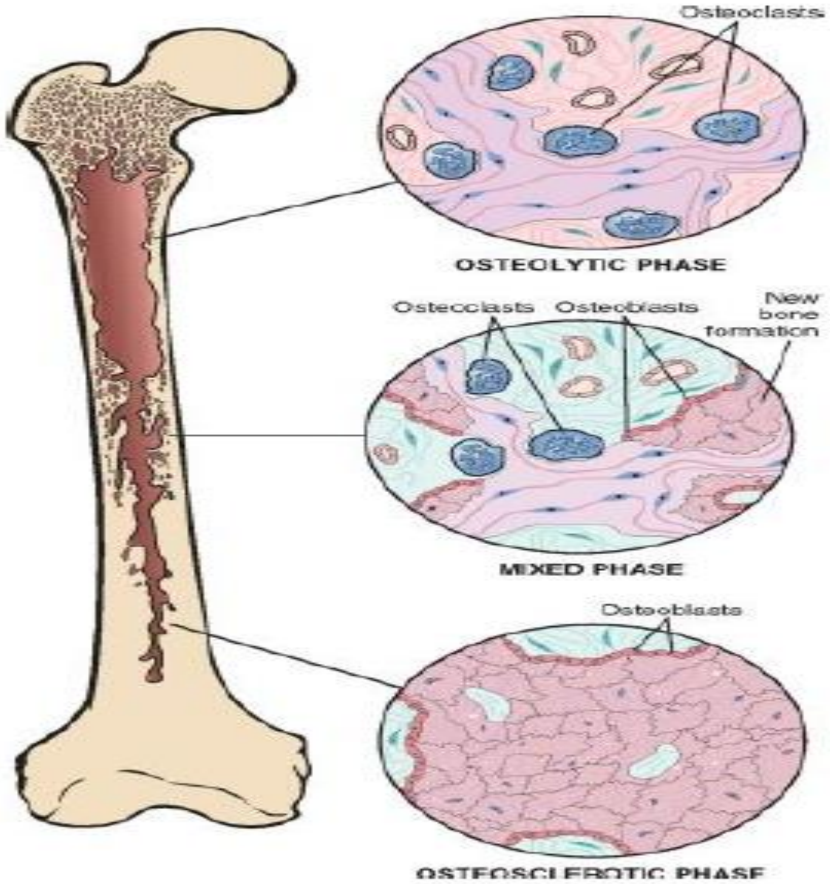
Paget's disease (osteitis deformans)

- There is excessive uncontrolled resorption of the bone by large multinucleated abnormal osteoclasts
- Excessive osteoclastic erosion in waves leads to localised destruction of trabecular and cortical bone
- Each wave of destruction is followed by a vigorous but uncoordinated osteoblastic response producing new osteoid producing a woven bone
- Can result in enlarged and misshapened bones – unrelated to the functional stress of the bone
- Followed by disorganized bone remodeling
- Causes affected bones to weaken resulting in pain, misshapened bone, fractures, arthritis

Cont'd

- Usually localised to only a few bones - Pelvis, femur, lower lumber vertebrae most commonly affected
- Onset in late adulthood
- More common in the white population and incidence differs between countries
- Divided into three phases:
  - Initial osteolytic stage
  - Mixed osteoclastic-osteoblastic stage with predominance of osteoblastic activity
  - Osteosclerotic stage
- Net effect is gain in bone mass which is disordered

Phases of paget's disease



Pathogenesis

- Exact cause of paget's disease of the bone is unknown but environmental factor such as infections (measles virus)
- And genetic factors which shows an autosomal dominant mutation of the SQSTM1 gene in 40 – 50 %
- The SQSTM mutations enhance NF-κβ activation by RANK signaling, leading to increased osteoclast activity and an increased susceptibility to the disease.

Clinical features

- Most cases are asymptomatic
- Depend on extent and site of disease
- Incidental findings on radiological picture
- Pain localized to affected bone
- Secondary osteoarthritis
- Kyphosis due to compression fractures
- Pelvic asymmetry, bowlegs – arthritis or joint pain
- Tumour and tumour-like conditions may develop – sarcoma, giant cell tumour

Diagnosis

- Increased levels of alkaline phosphatase
- Radiological findings –
  - thick coarsened cortex
  - Lytic lesion
- Bone biopsy
- To exclude malignancies that mimic PD of the bone
- Treatment include pain relievers and bisphosphonates
- Surgery
  - Correct bone deformities
  - Decompress an impinged nerve

Osteomalacia

- Disorder characterized by Defect in bone matrix mineralization
- Osteomalacia in children is called rickets
- In adults disease is milder
- Due to inadequate levels of phosphate and calcium
- 1,25 dihydroxy-vitamin D_3 leads to increased calcium in the GUT & ↑blood calcium → promotes mineralization of bone, ↑calcitonin inhibits bone resorption
- Most common cause is deficiency of vitamin D or disturbance in vitamin D metabolism
- This results in osteopenia and weakening of bones

Causes

- Inadequate dietary intake
- Inadequate body synthesis
- Malabsorption due to intestinal disease
- Renal disease (renal osteodystrophy)

Clinical features

- Diffuse joint and bone pain – due fractures in the cortical plate and trabecular bone
- Muscle weakness
- Difficulty walking
- Hypocalcemia – positive Chvostek sign
- Weak soft bones
- Easy fracturing of bones
- Bending of bones – bow legs in rickets

Biochemical findings

- Abnormally low serum concentration of Vit D
- Low serum and urinary calcium
- Low serum phosphate
- Elevated serum alkaline phosphatase
- Elevated parathyroid hormone – due to low calcium
- Technetium bone scan shows increased activity

Treatment

- According to underlying disease
- Weekly administration of vitamin D in nutritional deficiency
- Treatment by injection for malabsorption

INFECTIONS

Osteomyelitis

- Is the Inflammation of bone and marrow
- This almost always denotes infection
- Maybe a complication of any systemic infection but frequently manifests a primary solitary focus of disease
- Different types of microorganisms are responsible but most infections due to bacteria
- Pyogenic bacteria and mycobacteria are more frequent causes
- May involve any bone in the body but certain bones are affected more by certain organisms

Cont’d

- Microorganisms reach the bone by various routes
  - Hematogenous spread
  - Extension from a contiguous site
  - Direct implantation
- In children most cases are hematogenous
- Osteomyelitis is divided into three types:
  - Acute
  - Subacute
  - Chronic

Pyogenic osteomyelitis

- Pyogenic organisms causing hematogenous infections are due to bacteremia
- *Staphylococcus aureus* is the most common organism – up to 80%
- *E coli*, *Pseudomonas*, *Klebsiella* more frequent in IV drug users and people with UTI
- *Haemophilus influenzae* and *B group streptococci*
- *Salmonella* infections common in sickle cell disease
- No organisms isolated in 50% cases

Pathophysiology

- Pathologic change determined by age of patient, bone affected, type of organism, host factors
- Infection usually starts in metaphysis, permeates cortex and spreads through marrow
- Inflammatory reaction causes bone necrosis
- Dead bone (sequestrum) is surrounded by new bone (involucrum) and granulation tissue
- Infection continues as long as dead bone is present

Clinical features

- Fever
- Fatigue, malaise
- Irritability
- Restriction of movement of limb
- Local oedema, erythema and tenderness
- Complications can lead to thrombophlebitis, arthritis

Diagnosis

- Early diagnosis of acute osteomyelitis is critical
- Prompt antibiotic therapy may prevent bone necrosis
- Diagnosis of osteomyelitis is clinical
- Depends on clinical presentation and radiological findings
- In the first one week few specific features are present both clinically and radiologically

Lab findings

- Raised total white cell count with raised neutrophil count
- Elevated C reactive proteins
- ESR usually elevated
- Blood cultures positive in 50% cases in hematogenous osteomyelitis
- Aspiration and cultures positive in 25% cases

Radiological findings

- Earliest findings are soft tissue swelling
- Lytic lesions within 7 – 10 days
- Periosteal elevation and later thickening
- Osteopenia
- New bone apposition
- Sequestrum seen 3 – 8 weeks
- CT scan, MRI, US scan show suggestive features

Chronic osteomyelitis

- May result from acute infection when treatment is delayed or inadequate
- Occurs in weakened host defences or in extensive bone necrosis
- May present with a discharging sinus
- May be due to certain microorganisms such as mycobacteria
- Brodie's abscess – subacute osteomyelitis seen more common in children – intraosseous abscess

Complication

- Pathologic fracture
- Secondary amyloidosis
- Endocarditis
- Sepsis
- Squamous cell carcinoma in sinus tract
- Sarcoma in infected bone

treatment

- Antibiotic therapy main stay of treatment in acute osteomyelitis
- Prompt antibiotic therapy to prevent complications
- Surgery in chronic osteomyelitis – sequestrectomy and irrigation

Tuberculous osteomyelitis

- May spread from pulmonary or extrapulomnary infection
- Organisms usually blood borne
- Direct spread from lung infection into a rib
- Bone infection usually solitary
- Vertebrae affected in 40% cases - Pott disease, followed by knee and hip
- Infection breaks through intervertebral discs and may cause compression fracture, psoas abscess

Types of bone infections

- 1. Medullary
- 2. Superficial
- 3. Localized defect with stable bone
- 4. Diffuse infection with involvement of bone stability

TUMOURS

Osteochondroma

- Also called exostosis
- Arises from mutation in the EXOSTOSIN 1 & 2; Heparin Sulphate
- Benign cartilage capped tumour attached to underlying bone by a stalk
- Most common benign bone tumour
- Majority are solitary
- Most present in late adolescence and early adulthood
- Appear sessile or mushroom shaped on x-ray
- Size varies from 1 to 20 cm

Clinical features

- Common in males below 25yrs
- Slow-growing masses, mostly in the distal femur, proximal tibia
- May present with pain if they impinge on a nerve or soft tissue
- Commonly detected as an incidental finding on radiological examination
- They usually stop growing at the time of growth plate closure
- Rarely may give rise to a sarcoma
- Treatment is surgical excision in symptomatic cases

Osteoma

- Common in middle aged and older persons
- Occurs in flat bones of skull and face
- Hard, bony protuberance
- Slow-growing lesions of little clinical significance, usually asymptomatic
- Occur solitary but may be multiple in *Gardner syndrome*
- Composed of compact lamellar bone similar to cortical bone
- May cause obstruction of drainage of nasal sinus, impinge on brain or eye or produce cosmetic problems

Giant cell tumour

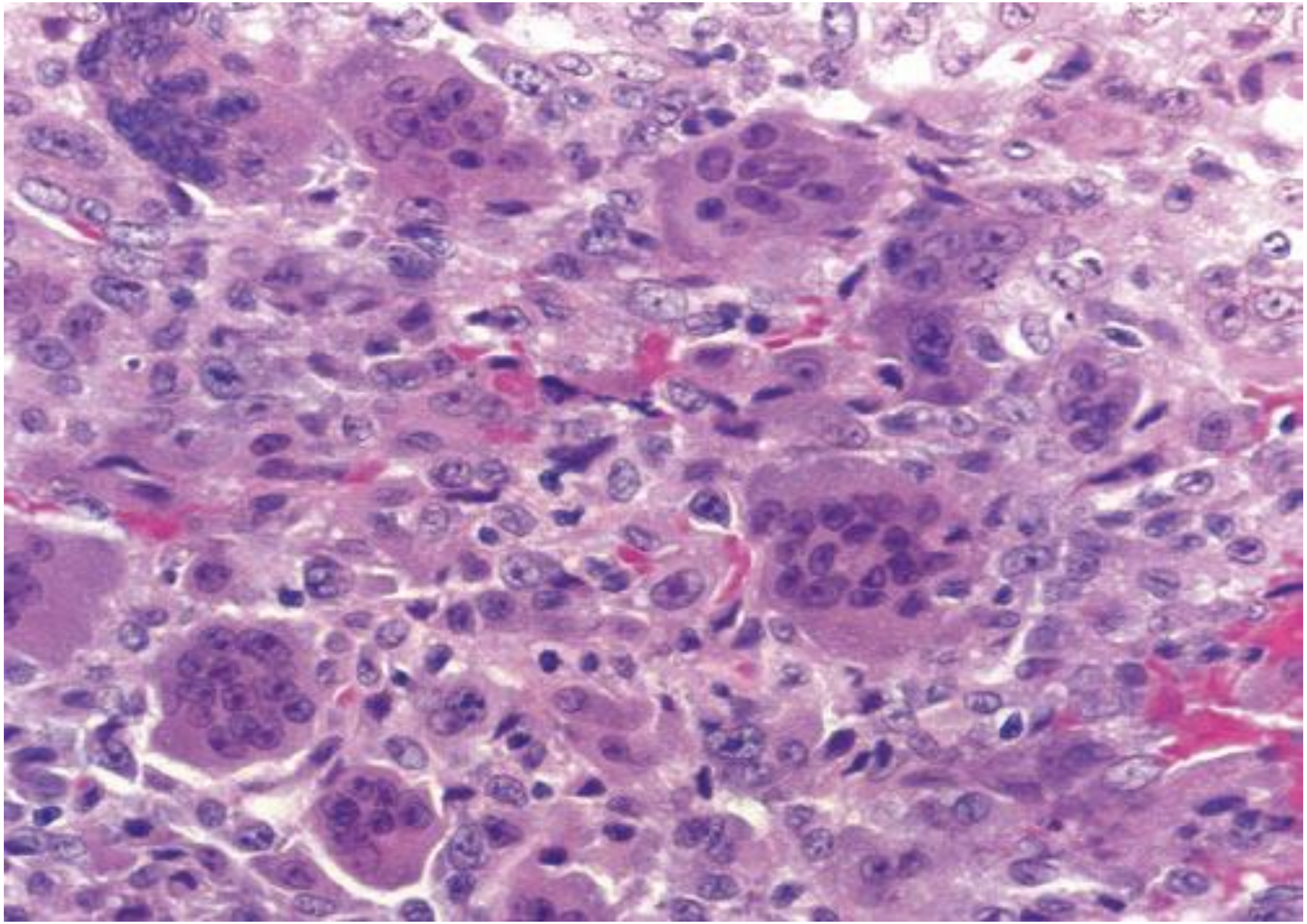
- Also known as osteoclastoma
- Benign locally aggressive tumour
- Tends to affect young adults between 20 - 40
- These tumours are extremely rare in children
- Sites commonly affected are distal femur, proximal tibia, proximal humerus, distal radius
- Involves the epiphyses with extension to end of bone

Clinical features

- Most patients present with bone pain
- May also present with a pathological fracture
- Rarely the tumour may be multicentric
- Radiologically – well defined lytic lesion involving epiphysis and metaphysis
- Typically lytic, expansile destructive lesion usually eccentric

Histological features

- Primary component are osteoclast-like giant cells and stromal cells
- Giant cells are distributed diffusely throughout the lesion
- Giant cells are not thought to be the neoplastic elements
- The stromal cells are the neoplastic cells and are mononuclear
- Giant cells are a result of fusion of circulating monocytes or histiocytes recruited into lesion
- Mitotic activity may be present in the stromal cells



Clinical course and treatment

- Tumour usually behaves in a benign fashion with an indolent course
- May be aggressive with uncontrolled local recurrences
- Treatment is complete surgical excision, may require bone grafting
- Radiation therapy for inoperable cases

Osteosarcoma

- Also known as osteogenic sarcoma
- Most common primary malignant mesenchymal bone tumour
- Usually affects children and young adults
- A bimodal distribution, second peak after 40 may occur in various existing bone condition such as Paget disease , M > F
- By definition a tumour in which osteoid is being synthesized by malignant cells
- Most occur in metaphysis of long bones
- Preferred sites are distal femur, proximal tibia, proximal humerus
- Typically initially involves medullary cavity then later grows into cortex

Cont’d

- Most tumours are unicentric
- Radiologically – may be totally opaque or radiolucent or both
- Rarely simulates a benign cyst of bone
- Codman’s triangle may be seen – due to elevation of periosteum by reactive new bone
- Most tumours will have extended into soft tissue by time of diagnosis
- May present with metastatic disease

Histological findings

- Tumour may be cystic, necrotic, hemorrhagic
- Osteoid or tumour bone produced directly by malignant cells
- Osteoid may be sparse or abundant surrounded by malignant stromal cells
- Osteoclast-like giant cells may be present
- Degree of anaplasia varies
- Different histologic subtypes are present

Clinical features and treatment

- Present as pain or swelling at the site
- May present as pathological fracture
- Sometimes presents as metastases – up to 20% patients present with metastases
- Traditional therapy consisted of amputation or disarticulation
- Limb sparing procedures are now available
- Pre-op or post-op adjuvant chemotherapy
- Surgical removal of metastatic lung nodule prolongs life

Ewing sarcoma

- Common in white boys less than 15 yrs
- Mostly occurs in the diaphysis of the long bones
- Has a onion appearance on X-ray
- Histologically, neuroectodermal originating small blue cells that resemble lymphocytes
- Associated with t(11,22) that leads to overexpression of EWS-FLI1 gene

Summary of the Notable Features of Selected Bone Tumors

Type	Description	Most Frequent Location	Incidence
Benign			
Osteochondroma (exostosis)	Cartilage-capped subperiosteal bony projection from bone surface	Lower end of the femur and upper end of the tibia	Males under age 25
Giant cell tumor	Tumor characterized by multinucleated giant cells and fibrous stroma	Epiphyses of the long bones, especially at the lower end of the femur and the upper end of the tibia	Females ages 20–40
Enchondroma	Intramedullary cartilaginous neoplasm	Bones of the hands and feet	No special age or sex incidence
Osteoma	Tumor of dense mature bone	Skull or facial bones; often protrudes into a paranasal sinus	Males of any age
Osteoid osteoma	Neoplastic proliferation of osteoid and fibrous tissue	Near the end of the diaphysis of the femur or tibia	Males under age 25
Osteoblastoma	—	Within vertebrae	—
Malignant			
Osteosarcoma	Osteoid formation is usually evident and is most obvious in the osteoblastic variant; may also produce cartilage (chondroblastic osteosarcoma), prominent fibrous background (fibroblastic osteosarcoma)	Tibia and femur near the knee	Males ages 10–20
Chondrosarcoma	Cartilaginous tumor	Pelvic bones, proximal and distal femur, proximal tibia, ribs, vertebrae	Males ages 30–60
Ewing sarcoma	Undifferentiated "small blue cell" tumor	Long bones, pelvis, scapulae, and ribs	Males under age 15

Bone fractures

- Is complete or partial break in the bone
- Caused by physical trauma are one of the most common abnormalities of the bone
- Degree can vary from simple crack in the cortical bone to a complex multiple fracture with fragmentation & displacement of bone pieces
- Severe damage to the surrounding soft tissues & sometimes exposure of the bone fragments to exterior through a large gaping wound open “compound” fractures

Causes of Bone fracture

- Severe physical trauma in mostly normal bone
- Trivial trauma may cause fracture when the underlying bone is abnormal → pathological fractures
- Osteoporosis (femur and vertebral column in the elderly)
- Vitamin D deficiency (small micro fractures without displacement)
- Paget's disease of the bone (structurally weak despite the increase in bulk)
- Primary tumours (giant cell tumour, myeloma)
- Metastatic tumours (metastatic carcinoma) → osteolytic metastases

Risk factors

Modifiable risk factors

- Low vit D
- Smoking
- Alcohol
- Glucocorticoid use

Non modifiable risk factors

- Increased age
- Congenital disorders
- Malabsorption problems

Types of fractures

- Classified based on the following 4 parameters

1. Position of the bone ends
2. Completeness of the break
3. Orientation of the break
4. Penetration of the skin

Cont’d

A. Position of the bone ends

- Non displaced (bones don’t move)
- Displaced (bone moves)

B. Completeness of the break

- Completely fractured
- Incompletely fractures

Cont’d

C. Orientation of the break

- Vertical axis (Linear axis)
- Horizontal axis (transverse axis)

D. Penetration of the skin

- Compound fracture (penetrates the skin)
- Simple fracture (do not penetrate the skin)

Pathophysiology

1. Inflammatory phase
2. Reparative phase
3. Remodelling phase

Signs & Symptoms

- Localised Pain
- Swelling
- bruising

Complications

Broken ends of the bone can damage surrounding tissues & structures

- Blood vessels →bleeding
- Nerves altered→ sensation
- Tears to the muscles or tendon

- Compartment syndrome
  - Bleeding or edema pressure inside limb →↓blood supply and tissue necrosis
- Fat embolism
- Malunion, delayed Union & Non-union
- DVT
- Pulmonary embolism

Diagnosis

- X-ray in two planes
  - Anteroposterior
  - Lateral
- MRI, CTS

The end!

Questions?